



PROBLEM STATEMENTS



Hackathon

at VIT Bhopal on 10th Feb

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1. SafeBrowse Junior: AI-Powered Safe Internet Companion for Children



• Concept:

A smart browser extension or mobile app designed for parents and educational institutions to protect children from harmful online content, including inappropriate websites, social engineering attempts, and excessive screen time. It provides real-time filtering, educational prompts, and AI-powered explanations (e.g., “This website contains adult content and has been blocked for your safety”).

• How to Build:

- Implement AI-based content classification using NLP to identify harmful or age-inappropriate sites.
- Integrate blocklists for categories such as violence, adult content, and malware (e.g., OpenDNS, Google Safe Browsing).
- Develop parental control dashboards and reporting features.
- Tech Stack: Hyperledger Fabric, IoT sensors, OpenAI for predictive analytics, Unreal Engine for VR.



2. Mobile Internet Safety: Privacy Pulse

- **Concept:** A mobile app that generates disposable, trackable aliases for every online account (emails, social media) and monitors them for breaches.
- **Unique Features:**
 - Dynamic Identity Cloaking: Creates unique pseudonyms linked to a user's real identity, which can be revoked instantly if compromised.
 - Breach Radar: Uses AI to scan the dark web for leaked aliases and alerts users in real time.
 - Privacy Score Dashboard: Rates apps based on data collection habits and suggests alternatives.
- **Tech Stack:** Zero Knowledge proofs for alias creation, TensorFlow Lite for breach detection, React Native.

3. Child Centric Cyber Safety: "Cyber Tales"

- **Concept:** A storytelling app where kids learn cybersecurity through interactive adventures, with parental controls and IoT integration.
- **Unique Features:**
 - Interactive Stories: Characters teach kids to spot phishing, manage screen time, and secure smart toys.
 - Parental Dashboard: Tracks children's app/device usage and blocks unsafe content using AI.
 - IoT Guardian: A physical "storybook" device that blocks insecure connections to home gadgets.
- **Tech Stack:** Flutter for cross platform apps, Raspberry Pi for IoT integration, GPT4 for dynamic story generation.



4. PhishGuard: Realtime Phishing Link Detector (Individual Safety)

- **Concept:** A browser extension or mobile app that scans URLs in real time (e.g., in messages, emails, or social media) and flags suspicious links.
- **How to Build:**
 - Use free APIs like Google Safe Browsing or VirusTotal.
 - Add a popup warning for flagged links.
- **MVP Features:** Basic URL scanning, color coded warnings (red/green), and a "Report Phish" button for crowdsourcing data.

5. SafeChat: Encrypted Panic Button for Institutions

- **Concept:** A secure chat app for schools/offices with a "panic button" to alert authorities during cyberbullying or threats.
- **How to Build:**
 - Use lightweight encryption (e.g., AES) for messages.
 - Integrate a panic button that sends geolocation + screenshots to preset contacts (e.g., IT team, police).
- **MVP Features:** End To End encryption, panic alert via SMS/email.



6. DataGuard: Mobile Data Usage Monitor (Individual Safety)



- **Concept:** An app that tracks background data usage and identifies apps leaking sensitive information (e.g., sending data to unknown servers).
- **How to Build:**
 - Use Android/iOS network usage APIs.
 - Flag apps with unusual data traffic (e.g., a calculator app sending GPS data).
- **MVP Features:** Traffic visualizations, alerts for suspicious activity.

7. CyberShield: AI-Powered Safe Browsing Assistant (Institutional)



- **Concept:** A Chrome/Firefox extension for schools/offices that blocks unsafe websites and explains risks in simple terms (e.g., "This site may steal passwords").
- **How to Build:**
 - Integrate AI APIs like OpenAI to generate plainlanguage risk explanations.
 - Use blocklists like OpenDNS.
- **MVP Features:** Site blocking, popup educational tips.



8. DeepShield: AI-Powered Deepfake Detection Tool



Concept:

A browser extension or mobile app designed to help users identify and report deepfake content in videos and images across social media platforms, news websites, and messaging apps. The tool educates users about manipulation techniques and provides real-time alerts (e.g., "This video shows signs of deepfake technology such as unnatural facial movements").

• How to Build:

- Utilize pre-trained AI models for image and video forensics (e.g., convolutional neural networks for face manipulation detection).
- Perform frame-by-frame analysis to detect artifacts, lighting inconsistencies, and blending errors.
- Implement metadata and audio signature checks for authenticity validation.

MVP Features:

- Real-time detection of deepfakes in media.
- User-friendly risk assessments and visual cues (e.g., probability of manipulation).
- Educational insights on how to recognize deepfakes independently.

9. CyberQuiz: Gamified Security Training (Institutional)



- **Concept:** A quiz app to teach cybersecurity basics (e.g., spotting phishing emails) with instant feedback and badges.
- **How to Build:**
 - Use React/Flutter for UI.
 - Create a JSON database of questions and answers.
- **MVP Features:** 10 quiz questions, scoring system, shareable badges.



10. SateNet: VPN with Built-in Threat



Intelligence

- **Concept:** A VPN service that integrates threat intelligence to block malicious websites and trackers while browsing.
- **Unique Features:**
 - Threat Blocking: Automatically blocks known malicious domains.
 - Tracker Prevention: Stops ad trackers and data collectors.
 - Speed Optimization: Uses AI to optimize server selection for faster speeds.
- **Tech Stack:** WireGuard protocol, AI-based server selection, threat intelligence APIs.



All the Best to All Candidates!



Go and build something great to contribute to society! 🌍 💡

the Future! 🔥